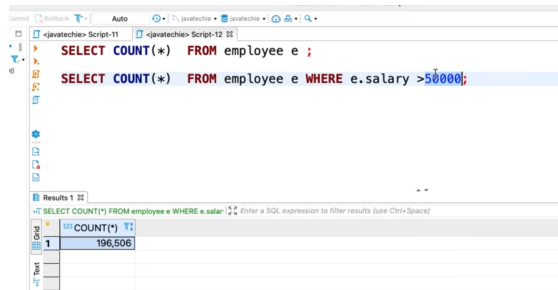
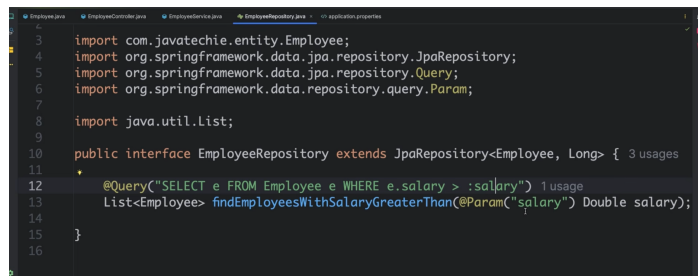


DB Query Optimization

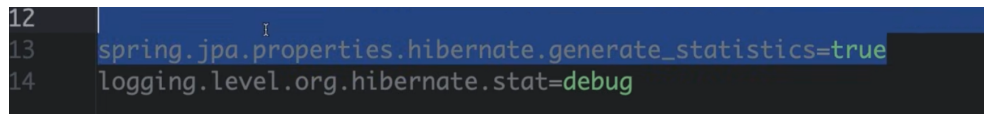
Query Hints – we have 1L+ records



Our repo snapshot



This property helps - statics of query.



You can see time taken for query execution



Just apply query hint – we are instructing hibernate behave like readOnly. We have so many such properties.



I can add many as shown below based on our requirement – google it.

```

11
12 public interface EmployeeRepository extends JpaRepository<Employee, Long> { 3 usages
13
14     @Query("SELECT e FROM Employee e WHERE e.salary > :salary") 1 usage
15     @QueryHints({
16         @QueryHint(name = "org.hibernate.readOnly",value = "true"),
17         @QueryHint(name = "org.hibernate.fetchSize",value = "50")
18     })
19     List<Employee> findEmployeesWithSalaryGreaterThan(@Param("salary") Double salary);
20
21 }
22

```

Enable hibernate 2nd level cache

```

public interface EmployeeRepository extends JpaRepository<Employee, Long> { 3 usages

    @Query("SELECT e FROM Employee e WHERE e.salary > :salary") 1 usage
    @QueryHints({
        @QueryHint(name = "org.hibernate.readOnly",value = "true"),
        @QueryHint(name = "org.hibernate.fetchSize",value = "50"),
        @QueryHint(name = "org.hibernate.cacheable",value = "true")
    })
    List<Employee> findEmployeesWithSalaryGreaterThan(@Param("salary") Double salary);

}

```

Force to retrieve data from cache.

```

@Query("SELECT e FROM Employee e WHERE e.salary > :salary") 1 usage
@QueryHints({
    @QueryHint(name = "org.hibernate.readOnly",value = "true"),
    @QueryHint(name = "org.hibernate.fetchSize",value = "50"),
    @QueryHint(name = "org.hibernate.cacheable",value = "true"),
    @QueryHint(name = "jakarta.persistence.cache.retrieveMode",value = "USE"),
    @QueryHint(name = "jakarta.persistence.cache.storeMode",value = "USE"),
    @QueryHint(name = "jakarta.persistence.query.timeout", value = "2000")
})
List<Employee> findEmployeesWithSalaryGreaterThan(@Param("salary") Double salary);

```

```

run JpaPerformanceApplication
100117: HQL: SELECT e FROM Employee e WHERE e.salary > :salary, time: 1029ms, rows: 196506
ion Metrics {

```